Genre-Controlled Short Story Generation Using QLoRA Adaptation

Sreekaran Reddy Ramasahayam
MS in Computer Science
University of Missouri Kansas City
Kansas City, Missouri
srz9k@umsystem.edu

Abstract* — Genre-controlled story generation requires a language model to produce coherent narratives while following a requested genre. This project explores whether QLoRA-based parameter-efficient fine-tuning can improve short story generation for fantasy, romance, and science fiction prompts using google/gemma-3-1b-it as the base model. The system was trained and evaluated on a curated dataset of 709 genre-labeled story examples, with the original pretrained model used as the base system and the QLoRA-adapted model used as the adapted system. Both systems were compared using the same task definition, prompt format, validation set, and evaluation metrics. The adapted model reduced validation loss from 3.82 to 3.31 and perplexity from 45.9 to 27.6, suggesting improved token-level fit to the genre-conditioned generation task. However, LLM-as-judge evaluation on sample outputs showed that the base model scored higher qualitatively on several story-quality dimensions. These results suggest that QLoRA improved token-level fit, but qualitative story quality still varied and remained limited by dataset size, compute constraints, and generation instability

Keywords— ***Genre-Controlled Generation, QLoRA, Gemma-3-1B, Parameter-Efficient Fine-Tuning, Short Story Generation, Natural Language Generation.***

I. Introduction

Story generation is a challenging natural language generation task because a model must do more than produce fluent text. It must also maintain narrative structure, follow the user's prompt, and preserve genre-specific features such as tone, setting, character behavior, and plot direction. In genre-controlled story generation, this challenge becomes more specific: the system must generate a story that is not only coherent, but also aligned with a requested genre such as fantasy, romance, or science fiction. This matters because controllable generation is important for creative writing assistants, interactive storytelling systems, and other applications where users expect the model to follow high-level creative constraints instead of producing generic text.

The task in this project is to generate short stories conditioned on a genre label and a story prompt. The project uses the WritingPrompts dataset from Reddit, accessed through Kaggle, based on the WritingPrompts corpus introduced by Fan et al. for neural story generation [1]. This project focuses on three genres: fantasy, romance, and science fiction. The base system is the original pretrained google/gemma-3-1b-it model before adaptation. The adapted system uses the same model with QLoRA-based parameter-efficient fine-tuning. This setup allows a fair comparison between the non-adapted model and the adapted model while keeping the task, prompt format, evaluation data, and metrics consistent.

This project is motivated by the practical challenge of adapting large language models under limited compute resources. Full fine-tuning can be expensive, especially in small research settings, so parameter-efficient methods such as LoRA and QLoRA provide a more accessible alternative for adapting language models under limited compute [2], [3]. Instead of updating all model parameters, QLoRA trains a small set of adapter parameters while using memory-efficient quantization. This makes it possible to adapt an instruction-tuned model to a specific creative generation task without requiring large-scale hardware.

The main contribution of this project is an end-to-end genre-controlled story generation pipeline that compares a base Gemma-3-1B model with a QLoRA-adapted version on the same validation setup. The final processed dataset contains 709 examples split into 567 training examples, 70 validation examples, and 72 test examples. Quantitatively, the adapted model reduced validation loss from 3.8268 to 3.3188 and perplexity from 45.9134 to 27.6261. However, qualitative evaluation showed mixed results: the adapted model improved token-level metrics, while the base model scored higher in the small LLM-as-judge sample. Overall, the project shows that QLoRA can improve model fit under limited compute, but better loss and perplexity do not automatically guarantee better creative writing quality.

II. Related work / Background

Prior work on neural story generation shows that producing long-form creative text is difficult because models must maintain coherence, character continuity, and narrative structure over multiple sentences. Fan et al. introduced the WritingPrompts dataset, a large collection of human-written stories paired with prompts from an online writing community, and used it for hierarchical story generation [1]. This project builds on that foundation by using WritingPrompts as the source corpus, but narrows the task to genre-controlled short story generation for fantasy, romance, and science fiction.

This project is also related to controllable text generation, where a model is expected to follow a user-specified condition such as topic, style, sentiment, or genre. Instead of training a story-generation model from scratch, this work adapts an existing instruction-tuned model to follow genre labels and prompts more consistently. The project therefore focuses on practical task adaptation rather than proposing a new model architecture.

For adaptation, this project uses LoRA and QLoRA, two parameter-efficient approaches designed to reduce the cost of adapting large pretrained models [2], [3]. LoRA freezes the pretrained model weights and trains small low-rank adapter matrices inside selected Transformer layers, reducing the number of trainable parameters compared with full fine-tuning [2]. QLoRA extends this idea by fine-tuning LoRA adapters through a frozen 4-bit quantized model using memory-saving techniques such as NF4 quantization, double quantization, and paged optimizers [3].

The base model in this work is google/gemma-3-1b-it, an instruction-tuned model from Google's Gemma 3 family of lightweight open models [4]. This project positions itself between earlier story-generation research and modern parameter-efficient LLM adaptation: it applies QLoRA to a small, genre-labeled subset of WritingPrompts and evaluates whether the adapted model improves over the same pretrained model before adaptation.

III. Data / Dataset / Corpus

The dataset for this project is based on the WritingPrompts dataset introduced by Fan et al. (2018), which contains prompt-story pairs collected from a Reddit creative writing community [1]. In this dataset, users respond to writing prompts with full stories, making it suitable for prompt-conditioned story generation. I initially explored multiple dataset options and considered building a custom dataset, but constructing a reliable story-generation corpus manually was impractical within the project timeline. The WritingPrompts dataset was selected because it is widely used in story-generation research, contains realistic human-written creative text, and is available through Kaggle. The full dataset contains approximately 272K prompt-story pairs.

For this project, I constructed a smaller genre-controlled subset focused on three genres: fantasy, romance, and science fiction. After additional filtering, the final processed dataset contains 709 examples. Each example was formatted for supervised fine-tuning so that the model receives a genre label and a story prompt, then learns to generate the corresponding story. The task can be summarized as: given a requested genre by me manually, and a prompt, generate a short story that follows the genre and remains relevant to the prompt.

The preprocessing stage cleaned and filtered the raw stories before training. Unicode text was normalized using NFKC normalization, <newline> markers and carriage returns were converted into standard newline formatting, and excessive whitespace was reduced while preserving paragraph breaks. Stories marked as deleted or removed were discarded. I also filtered out entries containing unwanted prompt artifacts or meta-writing language such as writing tags, critique requests, instruction markers, or formatting artifacts. To keep the dataset aligned with the target short-story generation task, additional filtering was applied to reduce noisy samples and improve the quality of the final subset.

After preprocessing, the final dataset was split into 567 training examples, 70 validation examples, and 72 test examples. The training set was used for QLoRA adaptation, the validation set was used to measure model performance during evaluation, and the held-out test set was reserved for final comparison and qualitative analysis. Both the base system and the adapted system were evaluated using the same task definition, prompt format, held-out examples, metrics, and evaluation procedure. This ensured that the comparison measured the effect of QLoRA adaptation rather than differences in data, prompting, or evaluation setup.

Because the dataset originates from Reddit-based creative writing data, there are important ethical and responsible-use considerations [1]. Although the data is publicly available through Kaggle, the stories may still contain social biases, informal language, sensitive themes, or stylistic patterns from the source community. The model may reproduce or amplify these patterns during generation. For this reason, the system should be treated as an academic prototype rather than a production-ready creative writing tool.

IV. Method

This project implements an end-to-end genre-controlled short story generation system. The system takes two inputs: a target genre and a story prompt. The target genres are fantasy, romance, and science fiction. Each input example is formatted as an instruction-style prompt that tells the model the requested genre and provides the story idea. The model then generates a short story conditioned on both the genre label and the prompt. The overall pipeline includes dataset preprocessing, prompt formatting, base model inference, QLoRA-based adaptation, evaluation, and a Streamlit demo interface.

The base model used in this project is google/gemma-3-1b-it, an instruction-tuned language model from the Gemma family [4]. I selected this model because it is small enough to experiment with under limited compute resources while still being capable of instruction-following text generation. The base system is the original pretrained Gemma-3-1B-it model before any task-specific adaptation. The adapted system uses the same model after QLoRA fine-tuning, which allows a fair comparison between the non-adapted and adapted versions of the same system.

The tokenizer was loaded from the same model checkpoint, google/gemma-3-1b-it, using the Hugging Face AutoTokenizer [4]. Using the matching tokenizer ensures that input formatting, tokenization behavior, and vocabulary remain consistent between the base model and the adapted model. The maximum sequence length was set to 1024 tokens. The prompt format was kept fixed across both systems so that the comparison measured the effect of QLoRA adaptation rather than changes in prompting.

For adaptation, I used QLoRA, a parameter-efficient fine-tuning approach that combines 4-bit quantization with LoRA adapters [2], [3]. The model was loaded using 4-bit NormalFloat quantization with double quantization through BitsAndBytes, and computation was performed using bfloat16 where supported [7]. Before training, the quantized model was prepared for k-bit training, gradient checkpointing was enabled, and caching was disabled to reduce memory usage. LoRA adapters were configured using Hugging Face PEFT and applied to the self-attention query and value projection layers by selecting modules whose names contained self_attn.q_proj or self_attn.v_proj [5].

The LoRA configuration used a rank of 16, alpha of 16, and dropout of 0.10 for causal language modeling. The model was trained using supervised fine-tuning with a batch size of 1, gradient accumulation of 4, learning rate of 5e-5, maximum sequence length of 1024, and the paged_adamw_32bit optimizer. Only the LoRA adapter parameters were trained, while the main pretrained model weights remained frozen.

V. Experimental Setup

The experimental setup was designed to compare the original base model and the QLoRA-adapted model as fairly as possible. The baseline system is the original pretrained google/gemma-3-1b-it model without any task-specific adaptation [4]. The adapted system uses the same base model after QLoRA fine-tuning on the genre-controlled story dataset [2], [3]. Both systems were evaluated on the same task: generating a short story from a given genre label and story prompt. The same prompt format, tokenizer, evaluation split, generation settings, and evaluation metrics were used for both systems so that the comparison focused only on the effect of QLoRA adaptation.

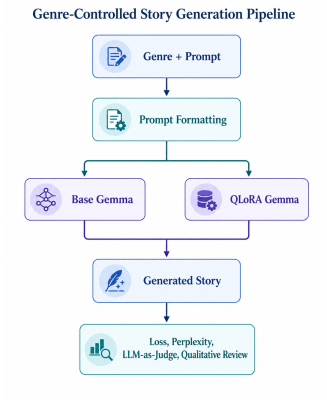
The dataset contained 709 processed examples across three genres: fantasy, romance, and science fiction. The split contained 567 training examples, 70 validation examples, and 72 test examples. The training set was used only for QLoRA adaptation. The validation and held-out evaluation examples were used to compare the base and adapted systems. The base model was not trained on this project dataset, while the adapted model was fine-tuned using QLoRA with LoRA adapters applied to the self-attention query and value projection modules.

The main quantitative metrics used were validation loss and perplexity. Validation loss measures how well the model predicts the target story tokens under the same evaluation setup, while perplexity provides an interpretable measure of language-model uncertainty based on the loss value. Lower values indicate better token-level fit to the evaluation data. Since this is a generation task, quantitative metrics alone do not fully capture story quality, so representative outputs were also reviewed qualitatively for genre alignment, prompt relevance, coherence, repetition, and ending quality.

In addition to manual qualitative review, I used an LLM-as-judge setup to evaluate six representative sample generations. Specifically, I used llama-3.1-8b-instant through the Groq API as a higher-capability evaluator [8]. The judge was given the original prompt, expected genre, generated story, and reference story as optional context, and was instructed to evaluate only the generated story using a structured rubric. The rubric included genre fidelity, prompt relevance, coherence, ending quality, repetition, style quality, format following, overall score, major issues, and short feedback.

The implementation was completed using Hugging Face Transformers, PEFT, TRL, PyTorch, BitsAndBytes, the Groq API, and Streamlit. Supervised fine-tuning was implemented using Hugging Face TRL's SFTTrainer [6]. The QLoRA model was trained with 4-bit NF4 quantization, double quantization, bfloat16 computation, gradient checkpointing, and the paged_adamw_32bit optimizer using BitsAndBytes quantization support [7]. Training logs, loss curves, generated samples, metrics tables, LLM-judge scores, and failure cases were saved as evaluation artifacts for reproducibility.

A Streamlit web application was also developed as the demo interface for the project [9]. The website allows a user to select a genre, enter a story prompt, generate outputs, and inspect the system behavior through an interactive interface. This demo component was used to show the end-to-end workflow more clearly.



VI. Results

This section presents the results of the base model and the QLoRA-adapted model for the genre-controlled short story generation task. Both systems were evaluated using the same task definition, prompt format, tokenizer, validation/test setup, and evaluation procedure. The base system was the original pretrained google/gemma-3-1b-it model, while the adapted system was the same model after QLoRA fine-tuning.

6.1 Quantitative Results

The main quantitative metrics used were validation loss and perplexity. Lower values indicate better token-level fit to the evaluation data. The QLoRA-adapted model improved over the base model on both quantitative metrics, but these metrics do not fully capture story quality.

System	Adaptation	ValLoss	Perplexity
Base Gemma-3-1B-it	None	3.82	45.9
QLoRA-adapted BaseModel	QLoRa	3.31	27.6

Validation loss decreased from 3.8268 to 3.3188, and perplexity decreased from 45.9134 to 27.6261. This indicates that the adapted model fit the genre-conditioned evaluation data better than the original pretrained model under the same evaluation setup. However, because story generation quality is not fully captured by token-level metrics, representative outputs and LLM-as-judge scores were also reviewed.

6.2 Training Curve

The QLoRA training curve showed a general decrease in training loss during fine-tuning. Training loss decreased from about 3.70 to about 3.21 across the training steps, while the final validation loss was approximately 3.29. This suggests that the model learned the supervised story-generation format, although the small dataset and short training run limit how much can be concluded from the curve alone.

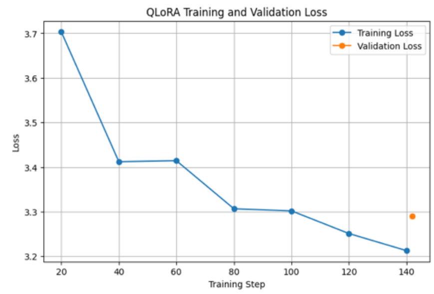


Fig. 1. QLoRA Training and Validation Loss Curve

6.3 Representative Base vs QLoRA Outputs

Because this is a generation task, validation loss and perplexity alone do not fully describe the quality of the generated stories. I compared representative outputs from the base model and the QLoRA-adapted model to check genre alignment, prompt relevance, and story structure.

Genre	**Prompt**	**Base Model Output**	**QLoRA Output**	**Observation**
Fantasy	A fantasy short story prompt …	"The air hung thick with the scent of lavender…"	"The air hung thick with the scent of lavender and …"	QLoRA gives a more concrete scene and character detail…
Sci-fi	A sci-fi short story prompt requiring a …	"The shimmer started subtly, a heat haze above Mars's…"	"The rain tasted like ash. It clung to my tongue …"	Base output has stronger sci-fi signals….

The representative outputs show that QLoRA improved structure and scene grounding in some cases, but the improvement was not consistent across all genres. In the sci-fi example, the base model included clearer genre-specific elements, while the QLoRA output drifted toward a more general dramatic style.

6.4 LLM-as-Judge Qualitative Evaluation

The LLM-as-judge evaluation used llama-3.1-8b-instant through the Groq API [8]. The judge scores were interpreted together with the quantitative metrics and manual review rather than as an absolute measure of quality.

Model	**Genre Fidelity**	**Prompt Relevance**	**Coherence**	**Ending Quality**	**No Repetition**	**Style Quality**	**Format Following**	**Overall Score**
Base	4.16	4.000	4.500	3.5	4.500	4.5	5.000	3.7
QLoRA	3.16	3.500	3.500	2.5	3.500	3.6	4.167	3.0

The LLM-as-judge results showed a different pattern from validation loss and perplexity. Although the QLoRA-adapted model achieved better token-level metrics, the base model received higher average judge scores across the evaluated sample outputs. This suggests that QLoRA improved token-level fit to the dataset but did not consistently improve perceived story quality in the small qualitative sample.

VII. Analysis

This section analyzes the generated outputs beyond the quantitative loss and perplexity results. Since story generation is a creative generation task, lower validation loss does not automatically mean every generated story is better. The results show a mixed outcome: QLoRA improved validation loss and perplexity, but the base model performed better on the small LLM-as-judge sample.

7.1 Representative Failure Cases and Difficult Examples

The main failure pattern was genre drift. In one sci-fi example, the base model included clear sci-fi elements such as Mars, ships, and alien imagery, while the QLoRA output became more atmospheric but less clearly science-fiction. This shows that the adapted model sometimes improved style but did not always preserve the requested genre.

Some romance outputs also showed weak genre alignment. Instead of focusing on emotional connection or relationship development, a few generations shifted toward unrelated dramatic or sci-fi-like settings. There were also minor formatting and completion issues, including title-like text, separators, meta-style wording, and weak endings.

7.2 What Worked

The main thing that worked was the QLoRA adaptation improving token-level fit to the dataset. Validation loss decreased from 3.8268 to 3.3188 and perplexity decreased from 45.9134 to 27.6261. QLoRA also made the project feasible under limited compute by training only adapter parameters instead of fully fine-tuning the entire model.

7.3 What Did Not Work

The adapted model did not consistently improve story quality. Some base model outputs were still better in terms of explicit genre signals, coherence, and ending quality. The LLM-as-judge scores also favored the base model in the small sample, showing that better loss and perplexity do not automatically translate into better creative writing.

7.4 Interpretation

One possible reason is that the reduced dataset size and short QLoRA training run helped the model learn surface-level patterns from the training data but did not fully improve higher-level story qualities such as coherence, genre consistency, and ending quality. The adapted model sometimes produced atmospheric writing, but it also showed genre drift, incomplete endings, and weaker prompt alignment in some examples.

VIII. Limitations and Responsible Use

This project has several limitations. First, the model was trained on a small processed subset of 709 examples, so it could not fully learn all genre patterns. Although QLoRA improved validation loss and perplexity, the generated stories still showed issues such as genre drift, repetition, weak endings, and occasional formatting artifacts.

A major limitation was compute availability. Due to GPU memory and RAM constraints, I could not increase the number of training epochs, use a longer maximum sequence length, or experiment with a stronger model. The training sequence length was limited to SFT_MAX_LENGTH = 1024, and generation was limited to around max_new_tokens = 300. These restrictions affected the model's ability to produce longer and more complete stories.

The QLoRA setup was also limited by compute. LoRA adapters were applied only to the query and value projection layers, specifically self_attn.q_proj and self_attn.v_proj. Adding more target modules may have improved adaptation quality, but it would have required more memory and training resources.

The evaluation also has limitations. Validation loss and perplexity measure model fit, but they do not fully capture story quality, creativity, genre fidelity, or emotional impact. I also used manual review and an LLM-as-judge setup with llama-3.1-8b-instant through the Groq API, but these judgments are still subjective and should be treated as supporting evidence rather than absolute truth [8].

Since the dataset comes from Reddit-based creative writing data available through Kaggle, it may contain biases, informal language, sensitive themes, or stereotypes from the source community [1]. The model can reproduce or amplify these patterns. Therefore, generated stories should be reviewed before public use. The system should not be used to generate harmful, deceptive, plagiarized, or impersonating content.

Overall, this system is an academic prototype for studying QLoRA-based genre-controlled story generation. It shows measurable improvement over the base model on token-level metrics, but it still requires human oversight and is not ready for unsupervised real-world deployment.

IX. Conclusion

This project explored whether QLoRA-based parameter-efficient fine-tuning can improve genre-controlled short story generation using google/gemma-3-1b-it. The main takeaway is that QLoRA improved token-level evaluation metrics while staying practical under limited compute. Compared with the base model, the QLoRA-adapted model reduced validation loss from 3.8268 to 3.3188 and perplexity from 45.9134 to 27.6261.

However, the qualitative evaluation showed a more mixed result. In the LLM-as-judge sample, the base model received higher average story-quality scores than the QLoRA-adapted model. This shows that better loss and perplexity do not automatically guarantee better creative writing quality. From this, I learned that creative generation needs both numerical evaluation and qualitative review.

Another key lesson was the importance of compute-aware design. Because of GPU memory and RAM limitations, I had to keep the model size, sequence length, generation length, number of epochs, and LoRA target modules limited. QLoRA made the project feasible, but the final quality was still shaped by these constraints.

For future work, I would increase the dataset size, train for more epochs, experiment with longer sequence lengths, and apply LoRA to additional modules beyond only the query and value projection layers. I would also test stronger base models, improve the genre-labeling process, and use a larger human or rubric-based evaluation set. Another next step would be improving the Streamlit demo so users can compare base and adapted outputs side by side and see evaluation feedback more clearly.

References

[1] A. Fan, M. Lewis, and Y. Dauphin, "Hierarchical Neural Story Generation," in Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics, 2018. [Online]. Available: https://aclanthology.org/P18-1082/

[2] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-Rank Adaptation of Large Language Models," arXiv preprint arXiv:2106.09685, 2021. [Online]. Available: https://arxiv.org/abs/2106.09685

[3] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient Finetuning of Quantized LLMs," in Advances in Neural Information Processing Systems, 2023. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2023/hash/1feb87871436031bdc0f2beaa62a049b-Abstract-Conference.html

[4] Google, "google/gemma-3-1b-it," Hugging Face Model Hub, 2025. [Online]. Available: https://huggingface.co/google/gemma-3-1b-it

[5] Hugging Face, "LoRA," PEFT Documentation. [Online]. Available: https://huggingface.co/docs/peft/package_reference/lora

[6] Hugging Face, "SFT Trainer," TRL Documentation. [Online]. Available: https://huggingface.co/docs/trl/en/sft_trainer

[7] Hugging Face, "BitsAndBytes Quantization," Transformers Documentation.[Online]. Available: https://huggingface.co/docs/transformers/quantization/bitsandbytes

[8] Groq, "Llama 3.1 8B Instant," Groq Documentation. [Online]. Available: https://console.groq.com/docs/model/llama-3.1-8b-instant

[9] Streamlit, "Streamlit Documentation." [Online]. Available: https://docs.streamlit.io/